laws, and safety protocols has restricted large-scale civilian deployment. Governments have often been cautious, balancing innovation with concerns over misuse, surveillance, and national security.

Equally significant are societal challenges, arising from public perception and ethical debates. Issues related to privacy invasion, job displacement, and the military use of autonomous drones have fueled global discussions. The acceptance of drones within urban environments, agriculture, or public safety continues to depend on public trust and transparent regulatory governance (Agrawal, et al. 2024).

7.1.2 Technological milestones driven by limitation

Each generation of drones reflects a direct response to the limitations of its predecessors.

- *First-generation drones* (pre-1970s) were constrained by primitive control mechanisms and short flight times, which led to improvements in remote-control and propulsion systems.
- *Second-generation drones* (1970s–1990s) faced challenges of navigation and communication, prompting innovations like GPS integration and data-link technology.
- *Third-generation drones* (1990s–2010s) encountered issues in autonomy and sensor accuracy, leading to the development of onboard computing and AI-assisted stabilization.
- *Fourth-generation drones* introduced semi-autonomous capabilities but struggled with endurance and cybersecurity, driving research into advanced materials, hybrid power sources, and encrypted communication.
- *Fifth-generation drones*, though technologically superior, face the most complex challenges: cybersecurity threats, cost, ethical